

BENTLEY C. HARRIS

First-Year Electrical Engineering Student · University of Queensland

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PROFILE

First-year Electrical Engineering student (Bachelor of Engineering (Honours) / Master of Engineering) with two years of formal fabrication and machining training and an ongoing portfolio of self-directed embedded electronics builds — Arduino-based control systems, sensor integration, and KiCad schematic and PCB design. White Card holder with a manual driver's licence. Available for regional and site-based placements, and comfortable working to procedure in hands-on, safety-critical environments.

EDUCATION

The University of Queensland — St Lucia, QLD 2026 – present

Bachelor of Engineering (Honours) / Master of Engineering — Electrical Engineering (first year)

Certificate II in Engineering — completed over two years during secondary school 2024 – 2025

MIG, stick and spot welding; oxy-acetylene cutting; metal and wood lathe operation; sheet metal fabrication in mild steel and aluminium; heat treatment; precision measurement and technical drafting.

St James Lutheran College Graduated 2025

- ATAR 93.85 (raw) — Dux of School, 2025.
- Represented the school at the 2025 Queensland Year 11/12 Mathematics Competition, Bundaberg.

TECHNICAL SKILLS

Electronics & Embedded Python programming, Arduino programming, sensor integration (IMU, RTC), motor controllers, MOSFET switching, buck converters, OLED displays, RF modules, battery and supercapacitor systems, soldering, perfboard construction, circuit fault diagnosis.

Design & Software KiCad schematic capture and PCB layout, CAD modelling for 3D printing, 3D printer operation, data logging and telemetry, spreadsheet data analysis.

Fabrication & Workshop MIG / stick / spot welding, oxy-acetylene torch, metal and wood lathe, drill press, band saw, angle grinder, belt sander, heat treating, sheet metal work.

Measurement & Drawing Precision measurement with calipers and scribes, engineering drawings and technical drafting, hand-drawn schematic documentation.

ENGINEERING PROJECTS

Full write-ups with photographs and build documentation at bentley-harris.github.io

Self-Stabilising Floating House — university project 2025 – 2026

- Designed and built a self-levelling floating platform driven by an Arduino control system with IMU feedback and a PID control loop — over 50 hours of design, build and iterative testing.
- Held a 1 kg shifting load stable against wind and water disturbance in a test tank. Tuning was never fully resolved, residual oscillation remained at the end of the project, and working out why taught me more than the build itself.
- Integrated motor control, power regulation and data logging to verify performance against measured results rather than assumption.

Embedded Electronics Portfolio — multiple builds ongoing

- Progressed from discrete sub-systems (motor and fan drivers, LED control, sensor interfacing) to complete builds combining sensing, control, power regulation and a user interface.
- Produced schematics and PCB layouts in KiCad and assembled hand-soldered perfboard circuits; diagnosed and repaired faults using systematic testing.
- Modelled and 3D-printed custom enclosures in CAD to house completed builds.

Workshop Fabrication Portfolio — Certificate II projects 2024 – 2025

- Completed a series of fabricated steel and aluminium pieces requiring welding, lathe work, sheet metal forming and heat treatment, each worked from my own engineering drawings.
- Built the habit of working to tolerance, checking dimensions with calipers before committing to a cut or weld.

SITE READINESS & CERTIFICATIONS

- Construction Induction Card (White Card) — current
- Queensland driver's licence — manual
- Responsible Service of Alcohol (RSA) Certificate
- Willing and available to work in regional and site-based locations for the full duration of a placement

WORK EXPERIENCE

Bartender & Waiter — Tres Salsas, Hervey Bay Sep 2024 – Feb 2026

- Worked front-of-house in a small team through sustained high-volume service periods, coordinating between bar, kitchen and floor to keep service moving.
- Maintained responsible service and food-safety standards under time pressure, without supervision during peak periods.
- Handled customer complaints and unexpected problems directly and resolved them on the spot.

Night Fill Team Member — Fresh & Save, Hervey Bay Nov 2023 – Nov 2024

- Completed stock replenishment to a fixed nightly deadline as part of a small crew, working to set procedures with minimal supervision.
- Held the role for twelve months alongside full-time study.

Casual Grounds Assistant — Hervey Bay Garden Centre 2+ years

- Maintained plant stock across the site on a recurring watering and care schedule, working independently to a consistent standard.

Work Experience Placement — Health Department — Fraser Coast Regional Council 2023

- Completed 45 hours across a local government health department, with exposure to regulatory inspection and compliance processes. Reference available on request.

LEADERSHIP & VOLUNTEERING

- Volunteer marker, UQ QAMT Mathematics Competition (2025).
- Senior Sports Leader, St James Lutheran College (2024) — organised and ran house sporting events.
- Debating Club member (2025, 8 months); Formal Planning Committee (2025); regular volunteer assistant for school events including GALA days.
- Selected participant, UQ InspireU Year 12 Camp (2025).

REFERENCES

Available on request.